

CAPSTONE PROJECT CHARTER

1. Project Information

Project Title: MetroPulse: NYC Taxi Demand Prediction and Batch Data Pipeline

Team Name: Team 06

Team Members:

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Supervisor: Assoc.Prof. Nguyen Ngoc Hoa

Start Date: 26/03/2026

Expected Completion Date: 01/06/2026

2. Project Background

Provide context and motivation for the project, including:

- Problem context: NYC taxi demand changes frequently due to weather and local events. Currently, historical TLC (Taxi and Limousine Commission) data is rarely combined with weather data for analysis.
- Importance of the project: Integrating these data sources allows us to forecast taxi demand. This helps fleet managers allocate vehicles more efficiently and reduces passenger wait times.
- Domain: Data Engineering, Big Data Architecture, and Predictive Analytics.

3. Problem Statement

The project aims to build an automated ETL pipeline that merges NYC TLC taxi records with external weather data to train a machine learning model for hourly demand forecasting.

4. Project Objectives

- Objective 1: Build a Batch ETL pipeline using Apache Spark to process 50 million rows of TLC Trip data (2023-2024) and integrate it with OpenWeather API data.
- Objective 2: Set up a PostgreSQL database to store the processed data for fast querying.
- Objective 3: Train an XGBoost model to predict hourly taxi demand per NYC zone.
- Objective 4: Develop a Streamlit web dashboard to display historical traffic trends and demand predictions.

5. Project Scope

In Scope:

- Processing 2023-2024 TLC Yellow Taxi data.
- Fetching and integrating OpenWeather API data.
- Batch data processing (daily/weekly updates).
- Predicting hourly ride demand per zone.

Out of Scope:

- Real-time data streaming (e.g., Kafka).
- Processing Uber/Lyft datasets.

6. Project Deliverables+Milestone

No	Deliverable	Milestone (date)	Description
1	Project Proposal	12/04/2026	Define scope, datasets, and objectives.
2	Project Specification (architecture, functionalities, non-functionalities, ...)	01/05/2026	System architecture, ERD, and API specifications.
5	Source Code + Readme	01/06 – 10/06	GitHub repo with Spark jobs, SQL scripts, and Dashboard code.
6	Final Report	01/06 – 10/06	Technical documentation and model evaluation metrics.
7	Presentation+Demo	01/06 – 10/06	Live demo of the pipeline and dashboard.

7. Roles and Responsibilities

Member	Role	Responsibilities
Member 1	Team Leader	Cloud deployment (AWS/Docker) and project documentation.
Member 2	Data Engineer	Writing PySpark scripts and configuring Airflow DAGs.
Member 3	Data Engineer	Data cleaning, validation, and Database Schema (PostgreSQL) design.
Member 4	Data Analyst/ BI	Writing SQL queries and building the Streamlit UI/UX.
Member 5	ML Engineer	Feature engineering and training the XGBoost prediction model.

8. Technology Stack

Programming Language: Python, SQL.

Data Processing: Apache Spark (PySpark), Pandas.

Orchestration: Apache Airflow

Database: PostgreSQL

Machine Learning: Scikit-learn, XGBoost

Visualization: Streamlit

Deployment: Docker, AWS (EC2, RDS)

Version Control: Git & GitHub

9. Risk Management

Risk	Impact	Mitigation
Out of Memory (OOM) during Spark processing.	High	Test data logic on a 1% sample first. Use proper data partitioning in PySpark.
Missing or null values from Weather API.	Medium	Use Try/Except blocks in code and fill nulls with historical averages.

10. Success Criteria

- **Data Pipeline:** The pipeline successfully merges TLC and Weather data. It must process 1 month of raw data (~3 million rows) in under 15 minutes.
- **Dashboard:** The web app is accessible online, and database queries load charts in under 3 seconds.
- **Machine Learning:** The prediction model's Mean Absolute Error (MAE) is strictly lower than a baseline model (e.g., using the historical average of the previous week).